

27 April 2026



University of Antwerp
| Product Development

Medical Design Innovation

Prof. dr. Regan Watts

Session Overview

This session will have four parts:

Part 1: Medical Device Design – Leveraging Hospitals for Inspiration

Part 2: “Work In-Progress” Report from the Design Coal-face

Part 3: Antwerp Design Factory – Transition from Idea to Market

Part 4: Tour of the Antwerp Design Factory and School of Product Development

Part 1: Medical Device Design – Leveraging Hospitals for Inspiration

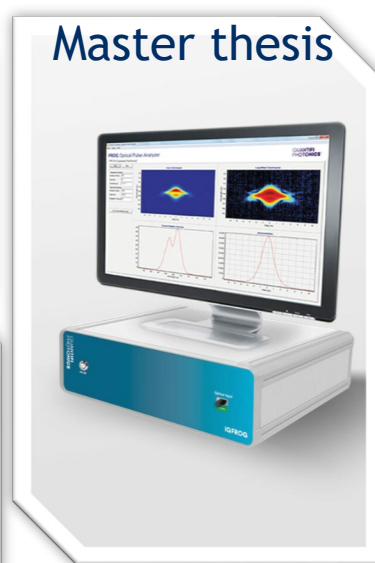
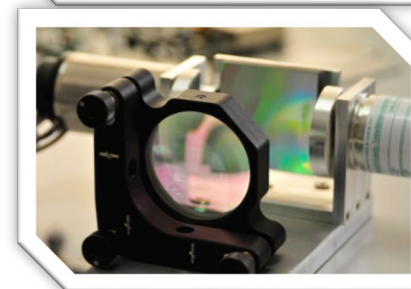
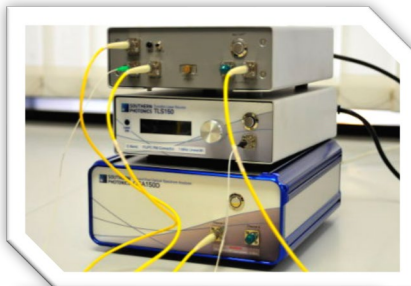
Who am I?

- University of Auckland, New Zealand
 - BTech (Optoelectronics), MSc (Physics) & PhD (Physics)
- Post-Doc at Dublin City University, Ireland (2010-2015)
 - Research projects in Photonics including FP7, ESA, EI
- Associate Prof. in Brunei Darussalam (2015-2017)
 - 'Engineering Design', Industrial Liaison
- Associate Prof. University of Antwerp (2017 - present)
 - 'Product and Engineering', Master programme
 - Research funding, typically proof-of-concept work



Products developed?

- Co-founder Southern Photonics (2001) spin-off from University of Auckland, New Zealand
 - USD\$ 2 million optical test-and-measurement market
- Daughter company 1, Quantifi Photonics, Series A, B, C venture funding
 - Acquired March 2025 by NASDAQ-listed Teradyne Inc, USD\$ 127 million



 SOUTHERN
PHOTONICS

QUANTIFI
PHOTONICS™

TERADYNE



Products developed?

- Daughter company 2, Southern Fiber Lasers
 - 10% purchased by Optical Engines (Boulder, Colorado, US)
 - Selective laser melting (SLM), additive manufacturing



Products developed?

- Co-founder Medimundi NV (2020), spin-off from UAntwerpen, design and manufacture FFP2/FFP3 respirator masks
- Patents led to multi-million dollar investment in production lines
- Upscale initiative to supply respirators to Belgian hospitals during the COVID-19 pandemic



Research Interests?

- Where I am now?
 - Routinely use 3D-printing platforms to create medical devices for functional proof-of-concept
 - Keep the users (patient, healthcare provider, biomed researchers) in the development loop
 - Validation of pre-production devices
 - Extended lifetime of medical devices

Where to focus?

- Medical Device Design extremely broad - where to (not to) focus?

Saint Francis Hospital (SFZ)

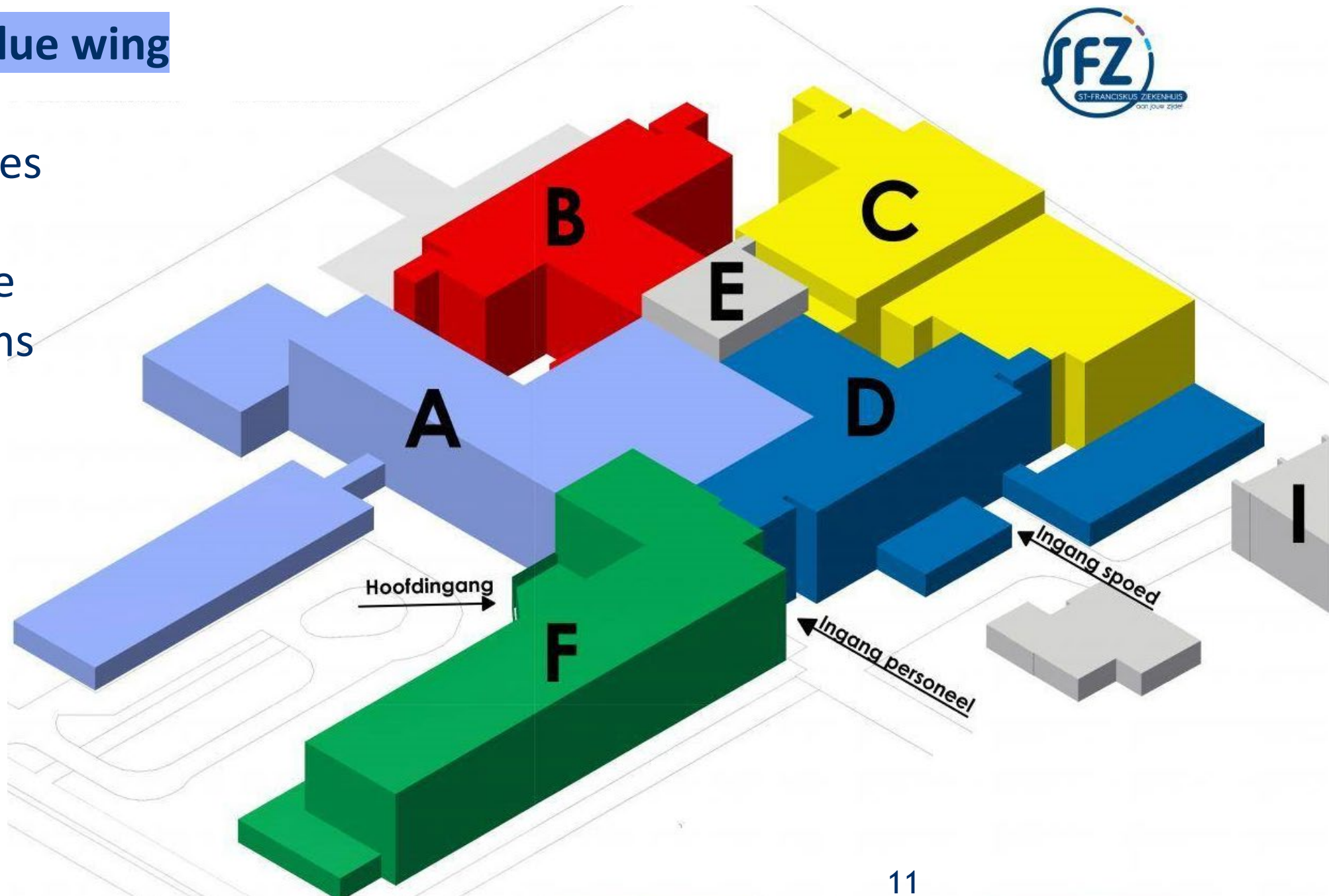
3550 Heusden-Zolder

Cardiology, Surgery, Geriatric care
Oncology, Pediatrics, Emergency
Obstetrics and gynecology



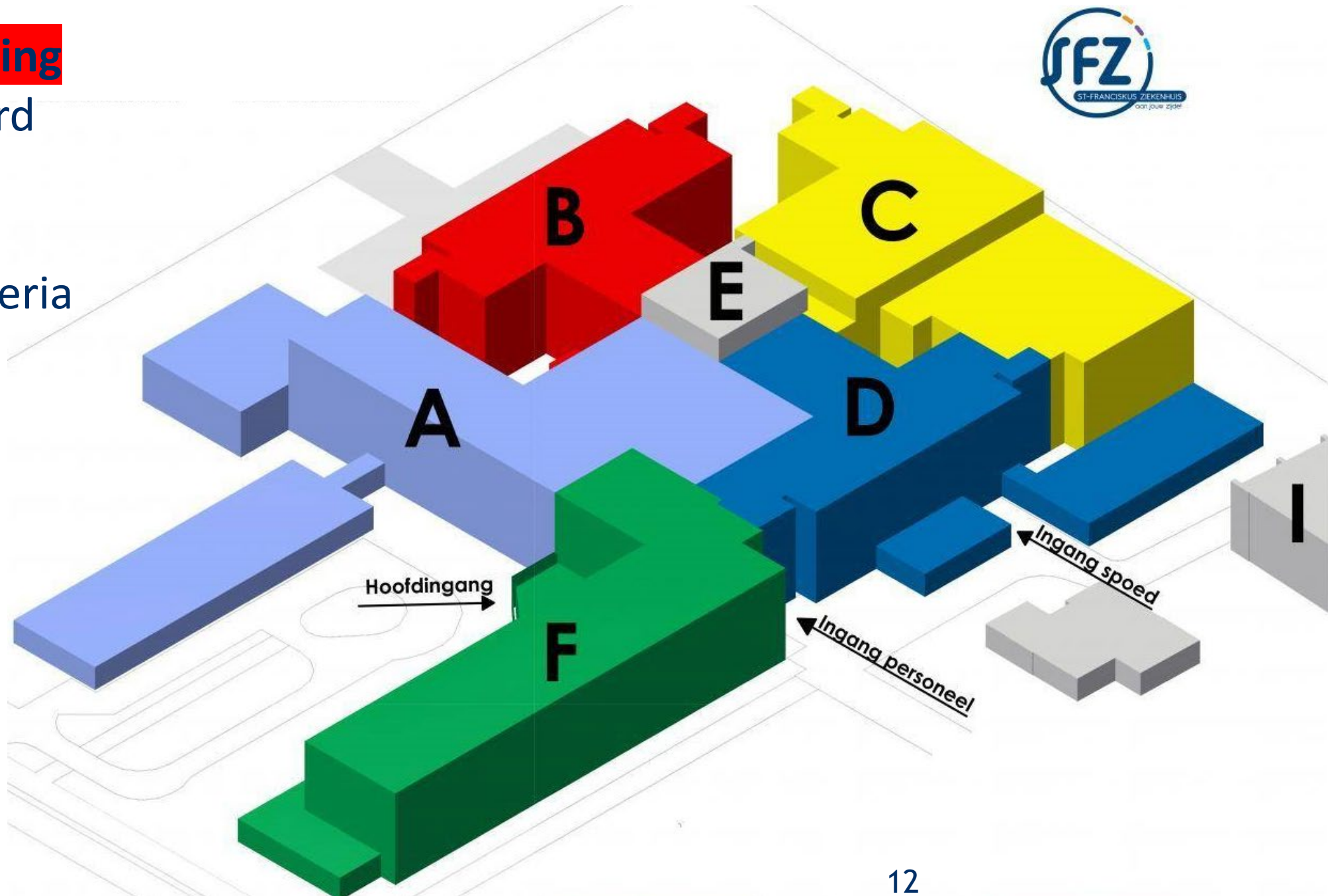
Block A – Light blue wing

- 3 Short stay
- 2 Internal diseases
- 1 Day hospital
- 0 Psychiatric care
- 1 Changing rooms



Block B – Red wing

- 3 Maternity ward
- 2 Geriatric care
- 1 Surgery
- 0 Kitchen/Cafeteria
- 1 Pharmacy/
Logistics



Block C – Yellow wing

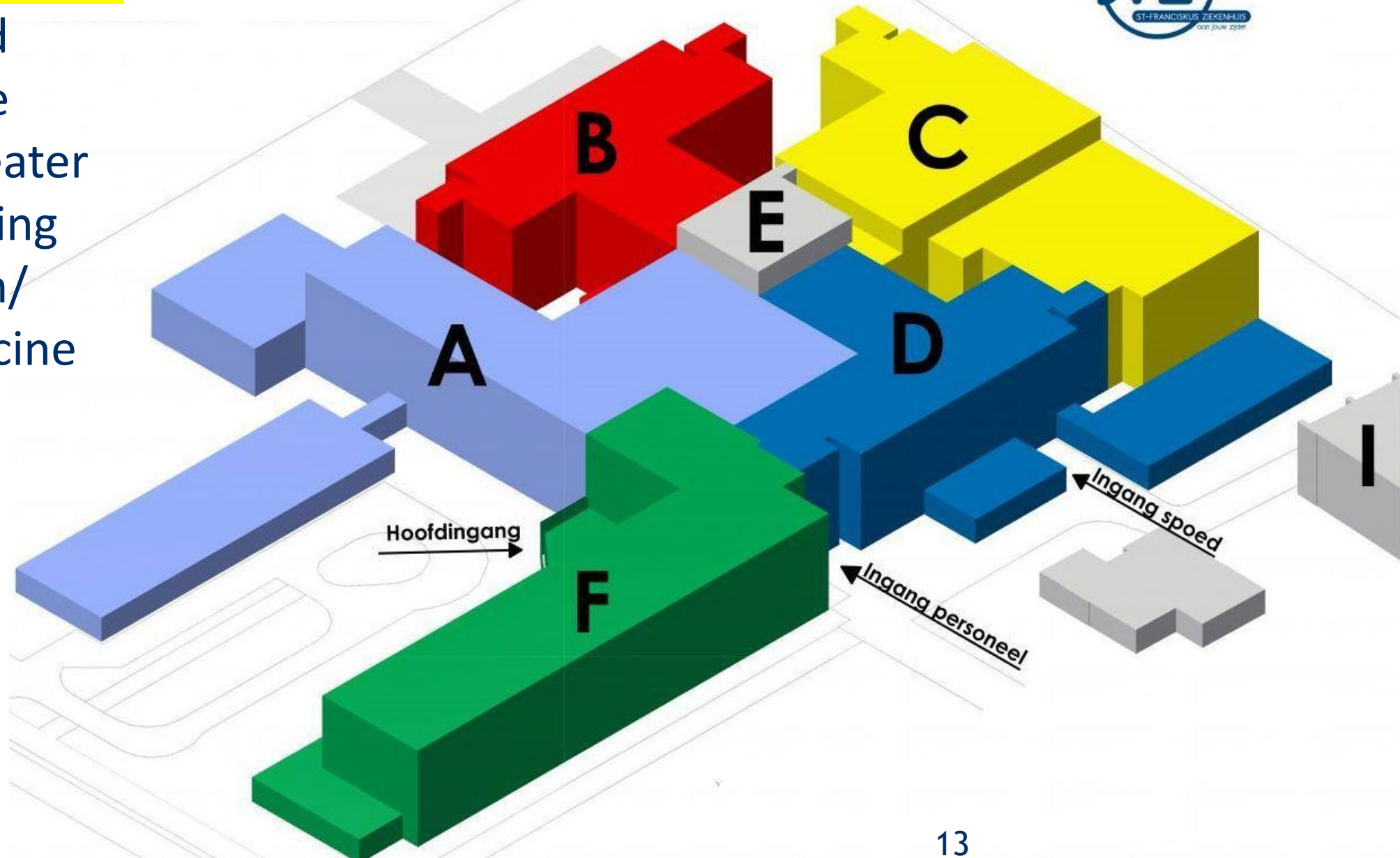
3 Delivery Ward

2 Intensive Care

1 Operating theater

0 Medical Imaging

-1 Rehabilitation/
Nuclear medicine



Block D – Dark blue wing

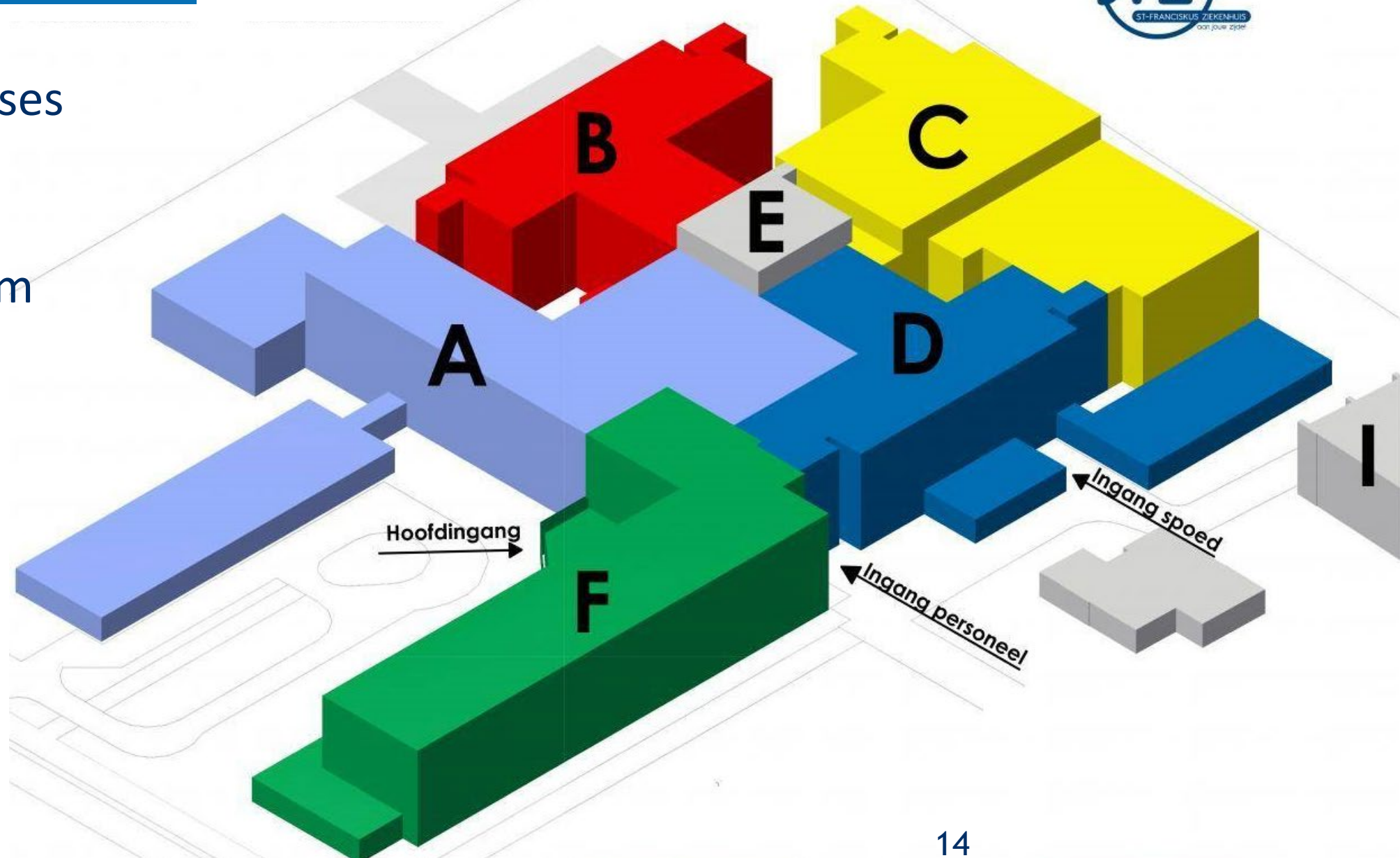
3 Pediatrics

2 Internal diseases

1 Surgery

0 Emergency

-1 Technical room



Block F – Green wing

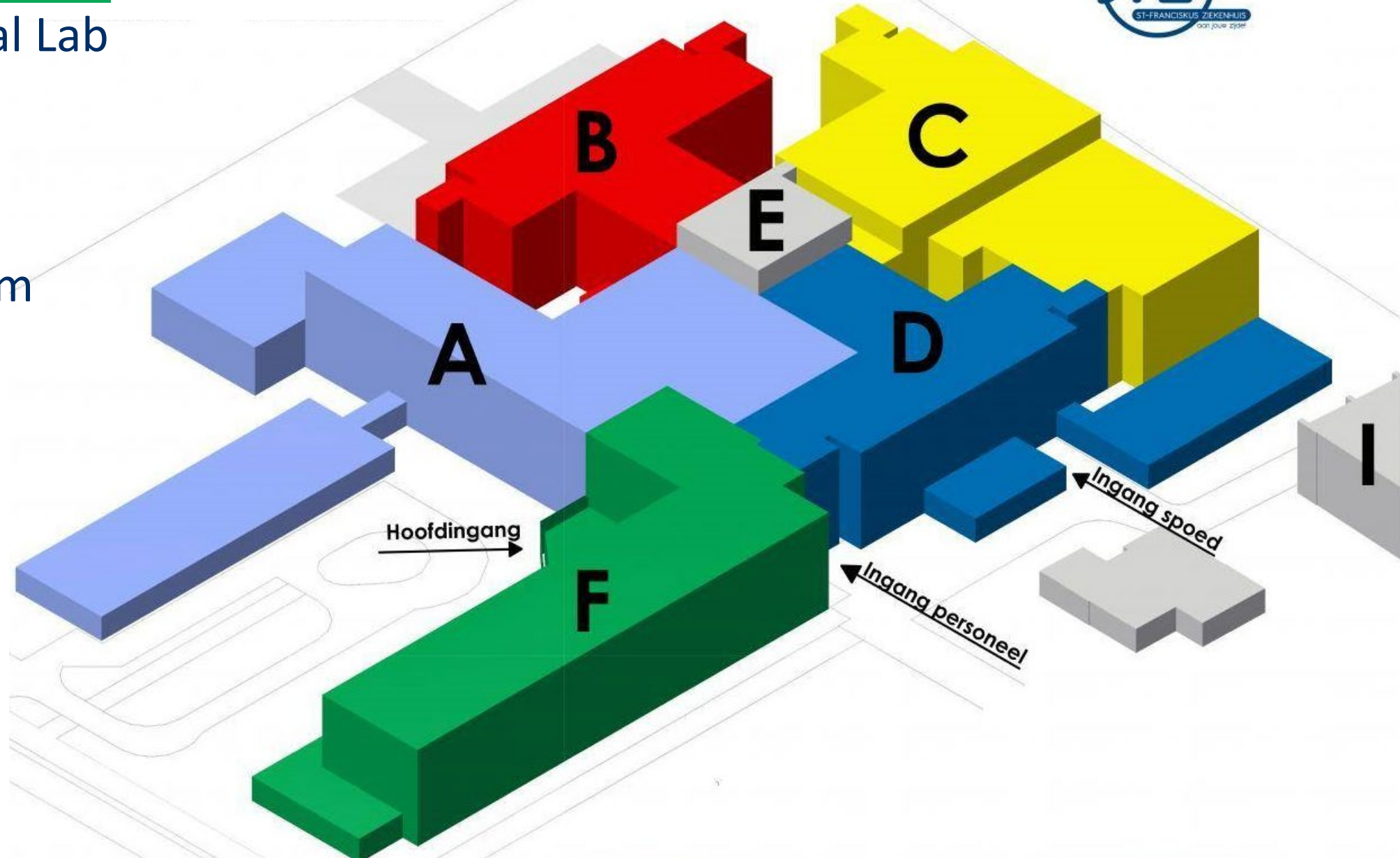
3 Admin, Clinical Lab

2 Polyclinic

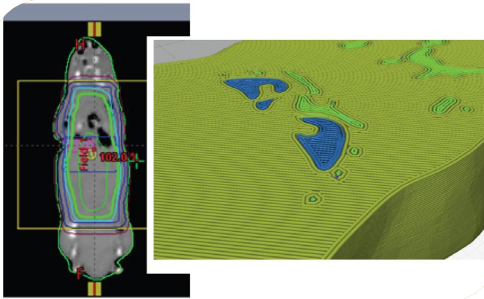
1 Geriatric care

0 Polyclinic

-1 Technical room



Floor -1



Mouse Phantoms
(FFF)

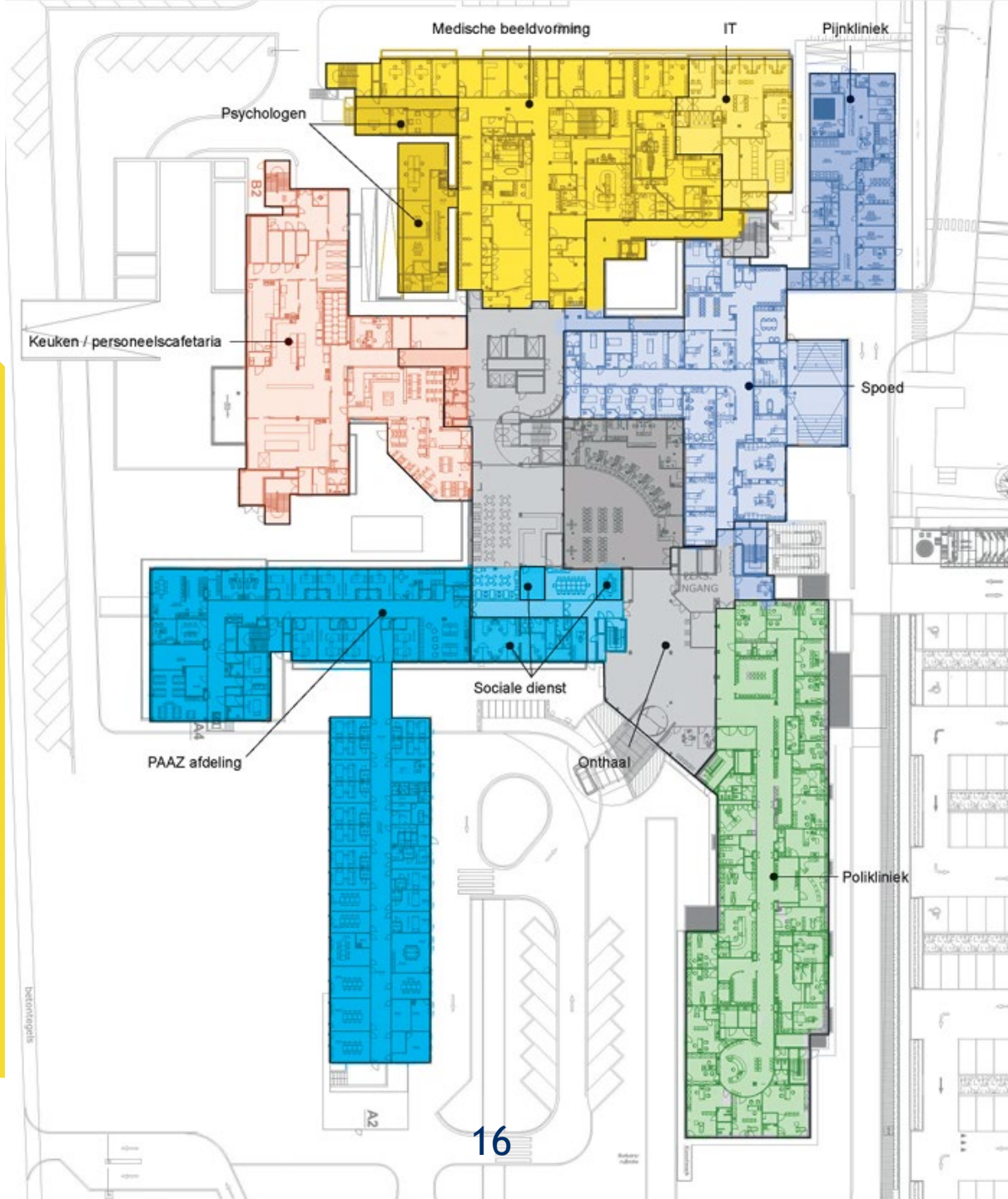


Helicus - Fluor-18

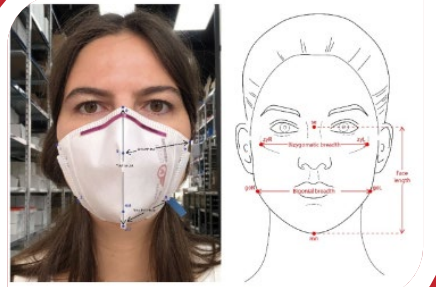


Flexible Ankle Foot Orthosis
(HP MJF, FFF)

Nuclear Medicine
Revalidation



Floor -1



Personalisation masks

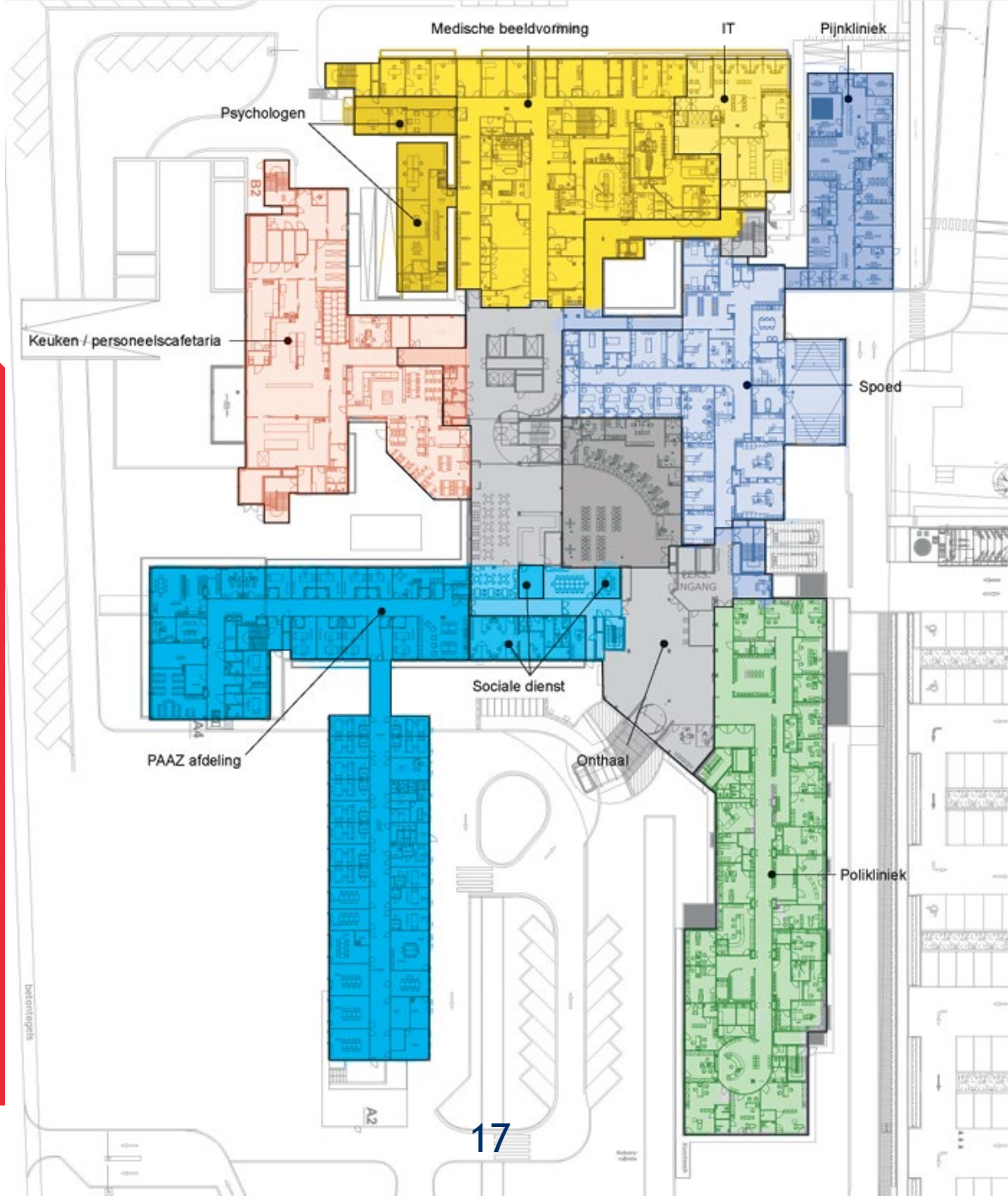


COVID-19 Vaccine Trays



Logistics - masks

Pharmacy
Logistics



Floor 0



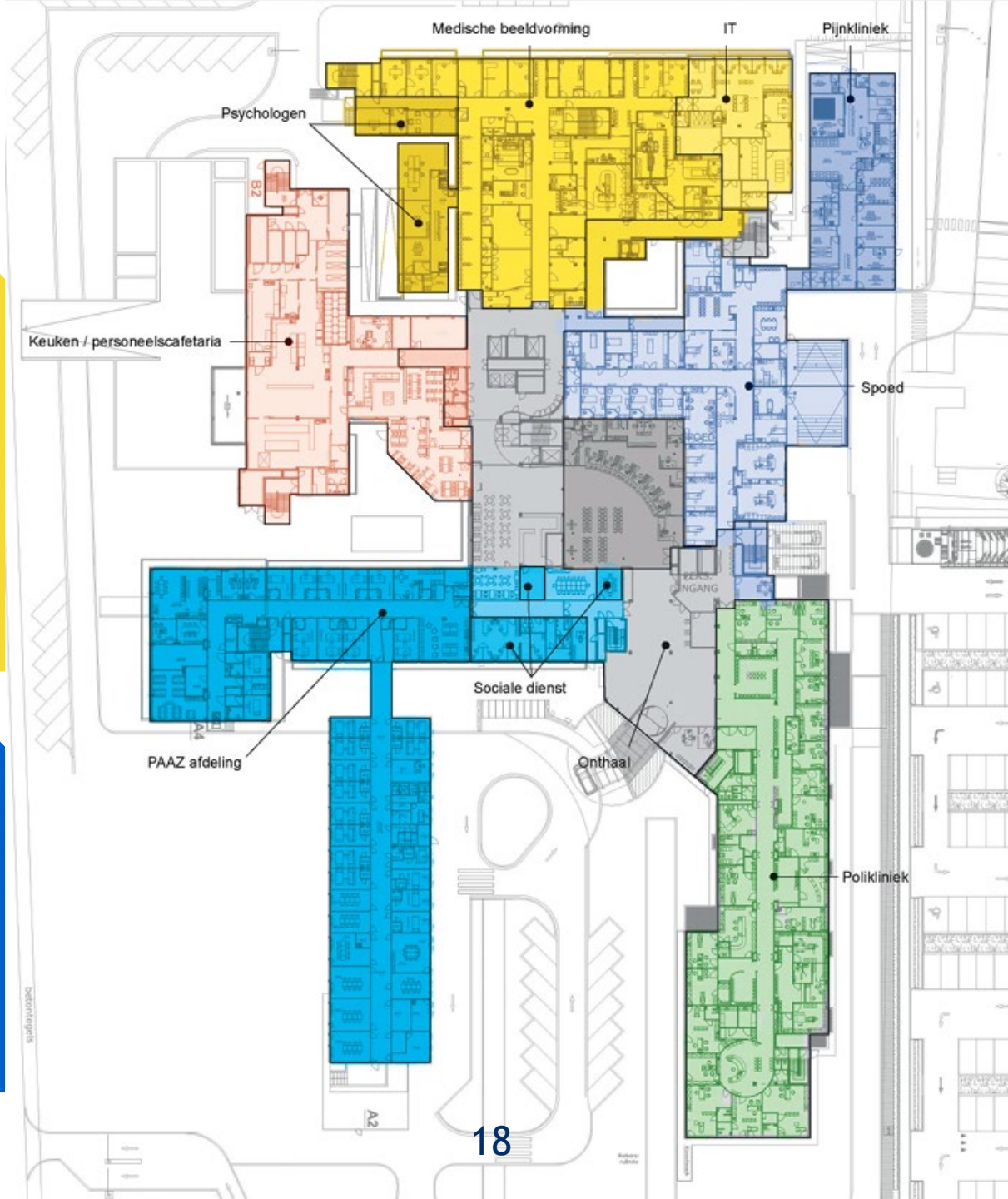
Ultrasound imaging -
Intubation

Medical
Imaging



Reuseable Suture Kits

Emergency



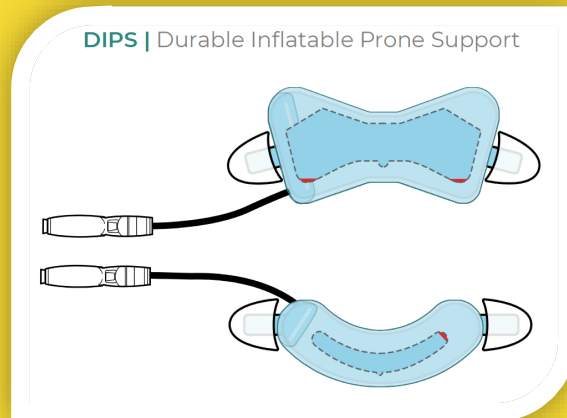
Floor 1



Reusable surgical gowns

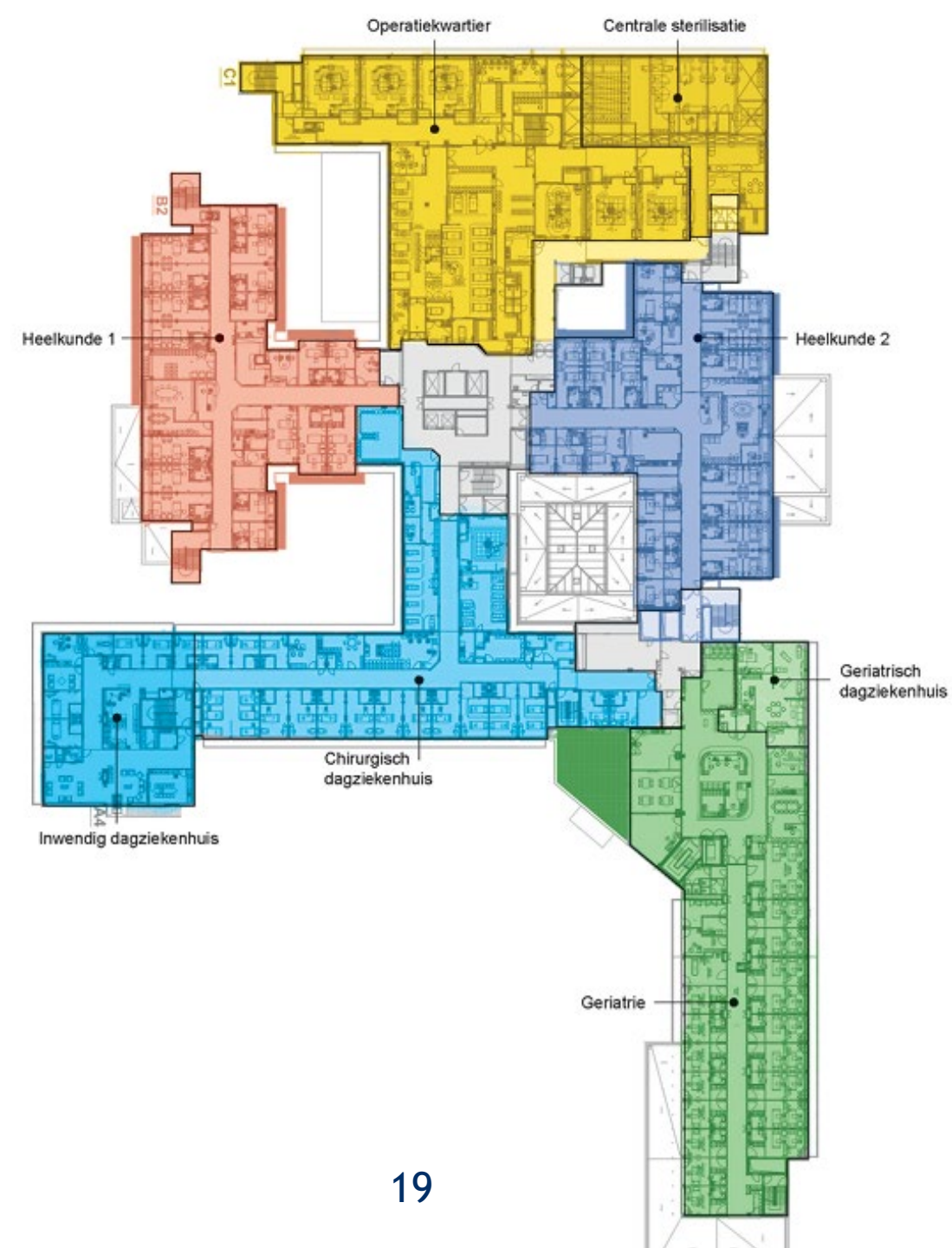


Helicus - Organ Transport

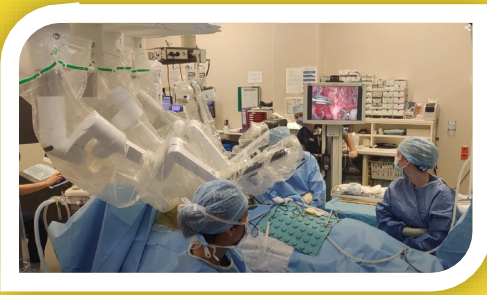


Ergotrics - Dr. Paul Depauw, Inflatable surgical cushions

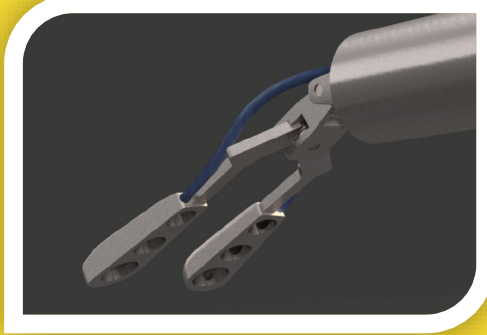
Operating
Theater



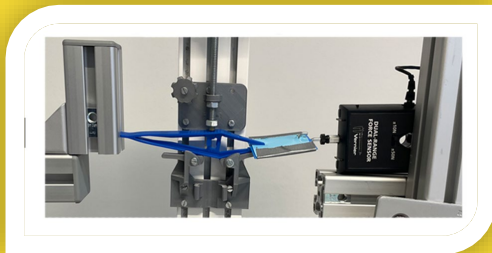
Floor 1



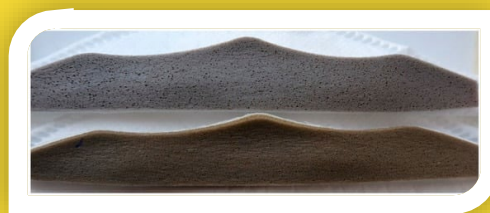
Prof. dr. Jeroen Hendriks -
Sympathetic nervous system



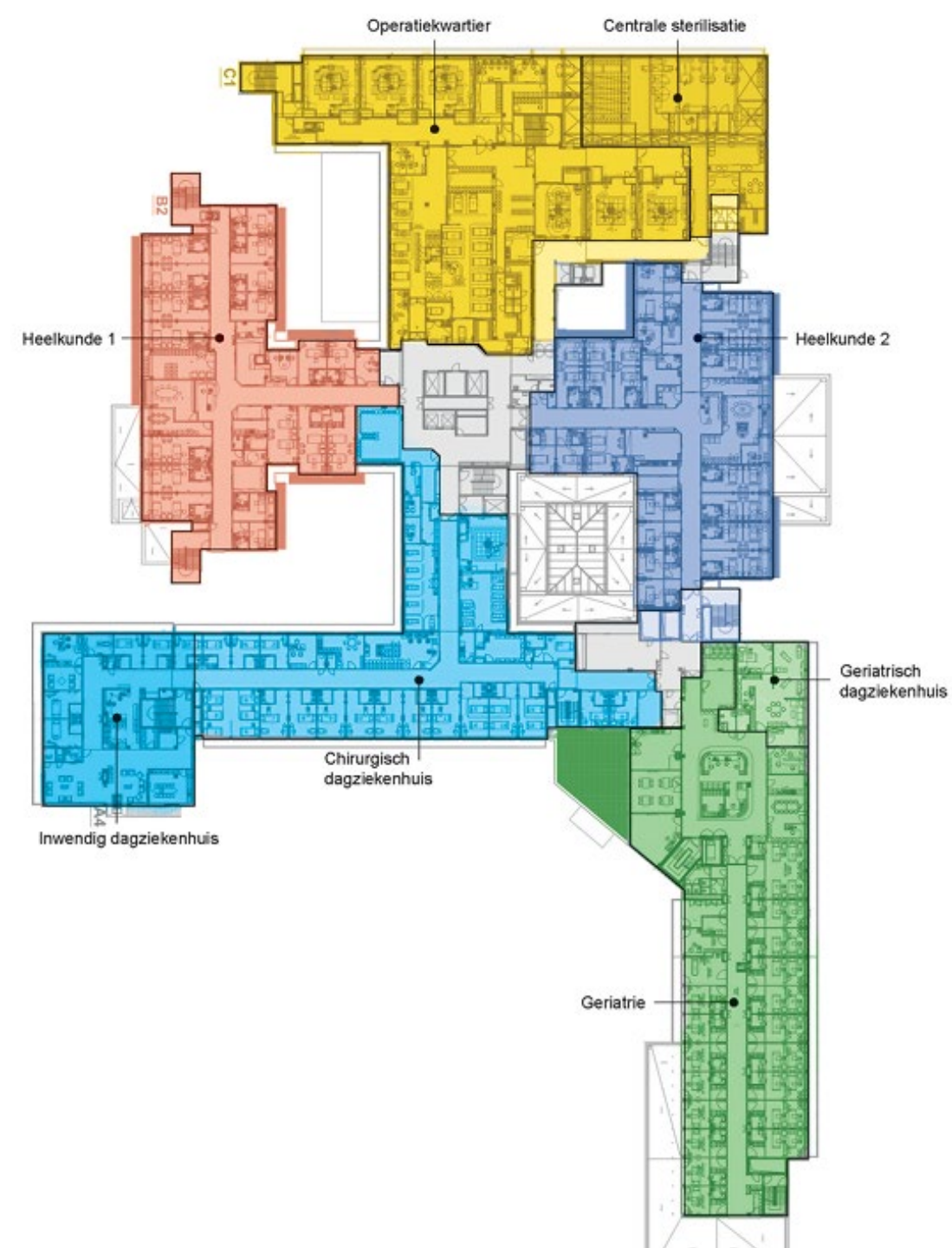
Robotic CABG



Autoclave



CIO₂
Operating
Theater
Central Sterilisation

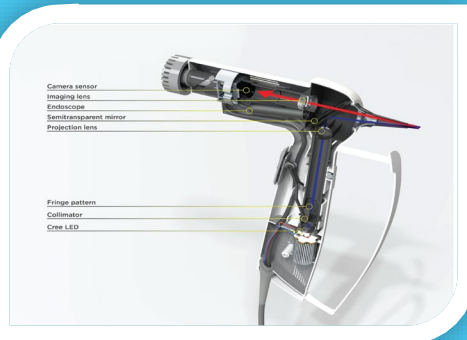


Floor 1



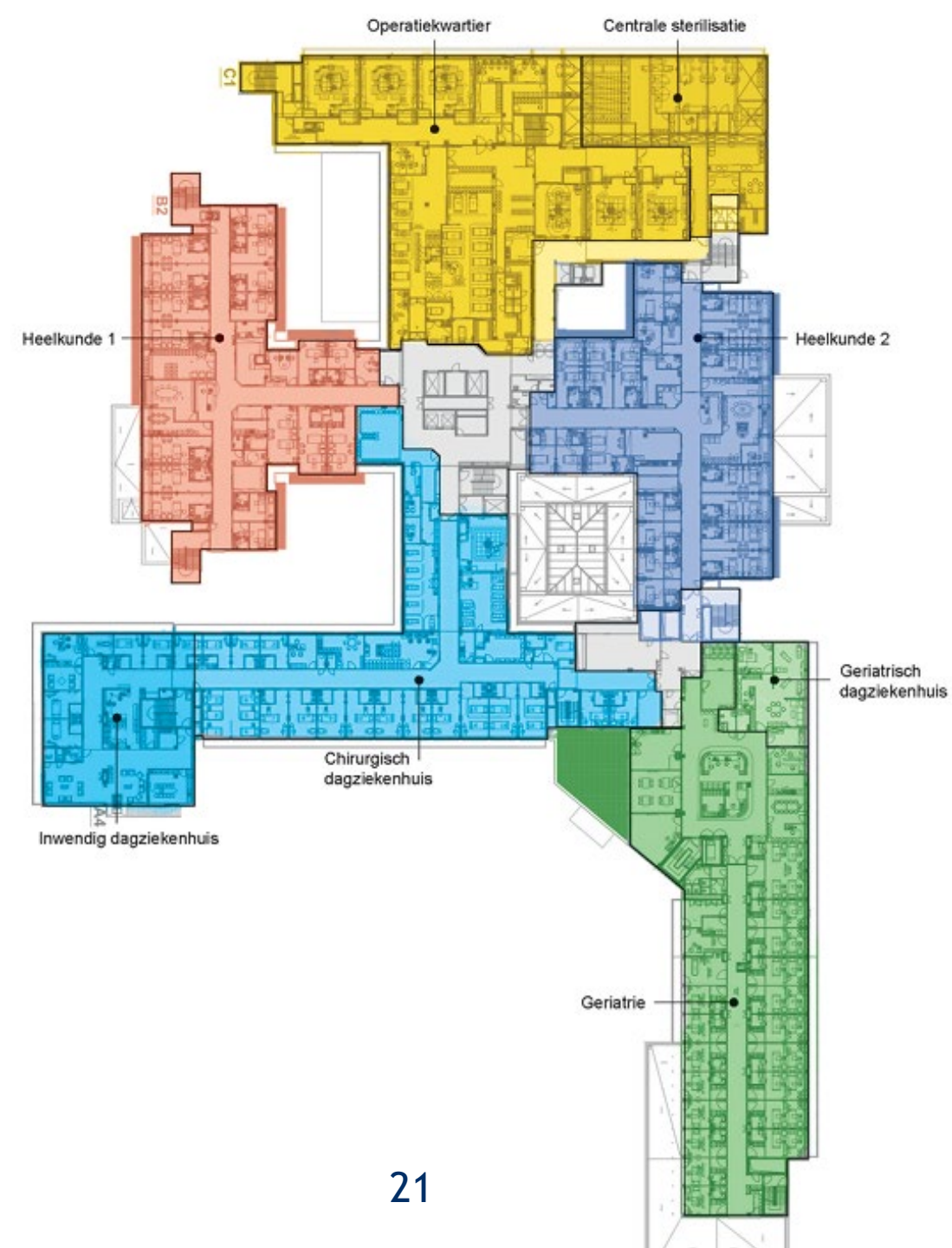
Smart potty-training

Urology

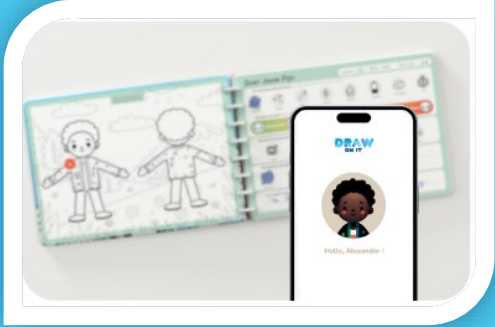


3D-Tympanoscope
(FFF)

Ear-Nose-Throat



Floor 2



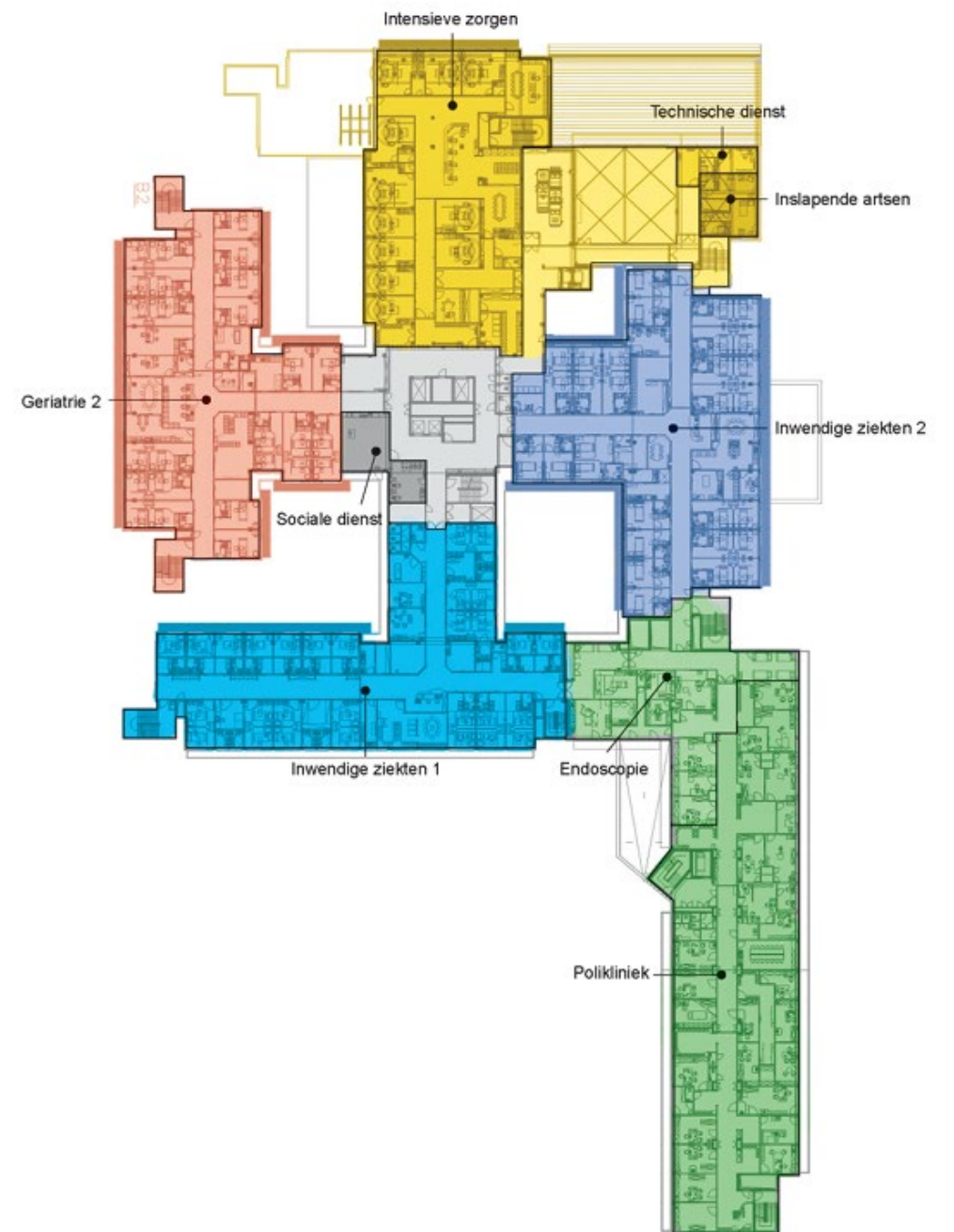
Sickle-cell anemia

Endocrinology



Continuous Glucose Monitors
(CGM)

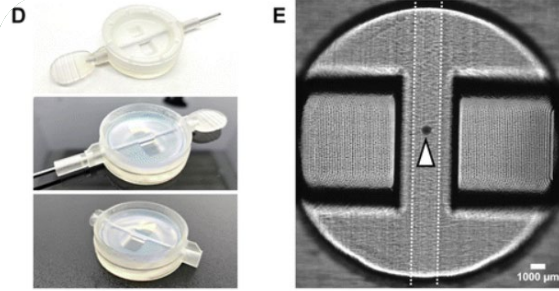
Diabetes



Floor 3



Glove box clean room



Fluidic-cell for Organoids

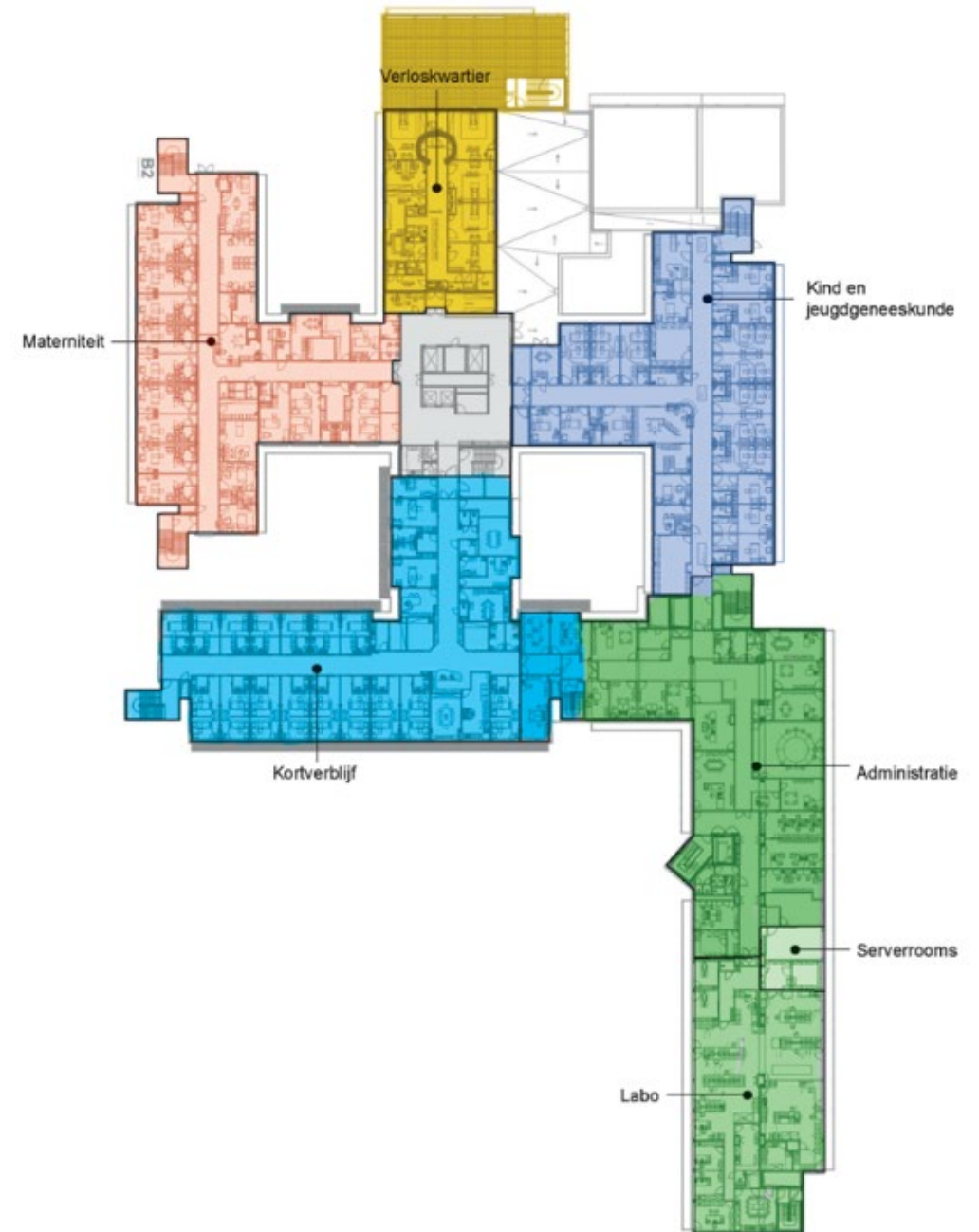


Free Bilirubin
(FFF, Polyjet)



Blood-Brain Barrier
(FFF, Polyjet)

Clinical
Lab



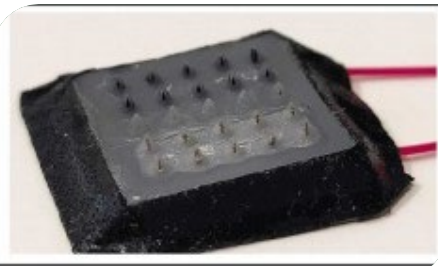
Floor 3



Sample prep for
Epidemiology



Phenylalanine sensor
(FFF)

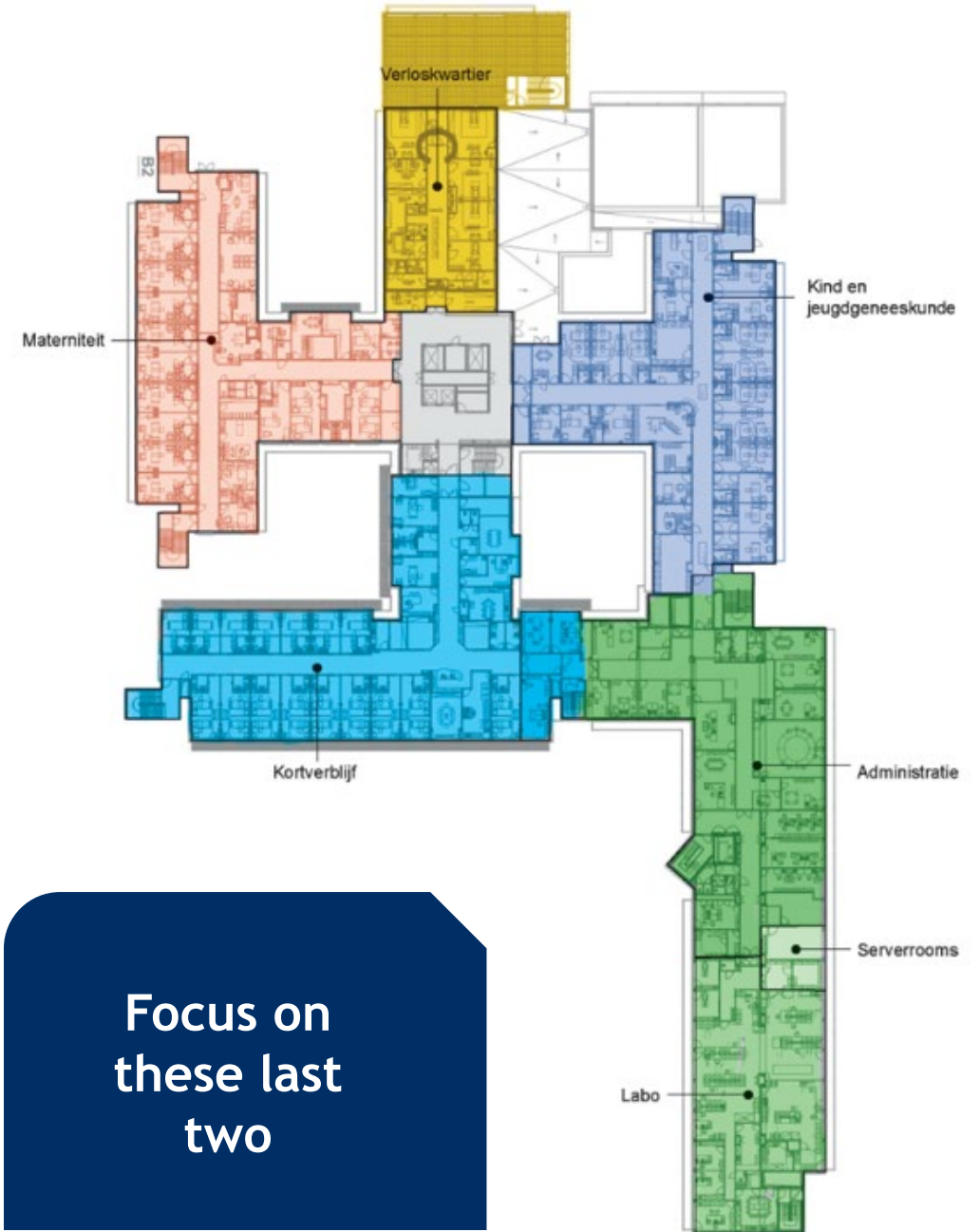


3D Printed
Microneedle Sensors
(mSLA)



Microneedle sensor for
immunosuppressants

Clinical
Lab



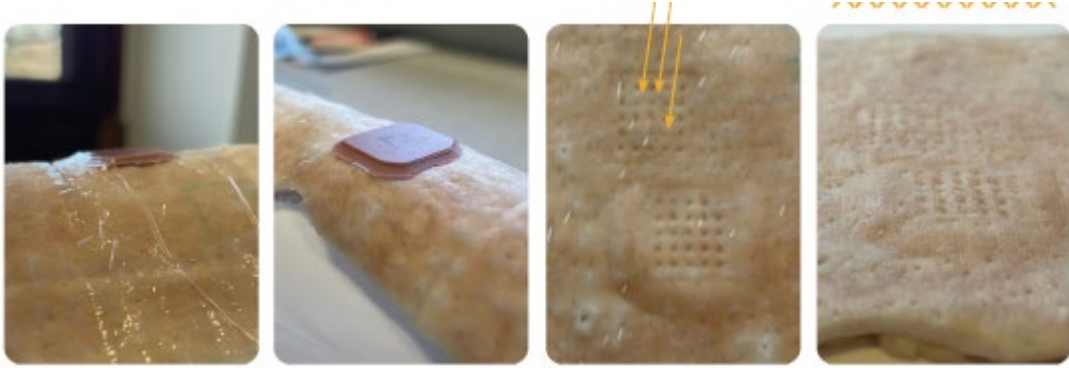
Focus on
these last
two

How do we go from concept to trial?

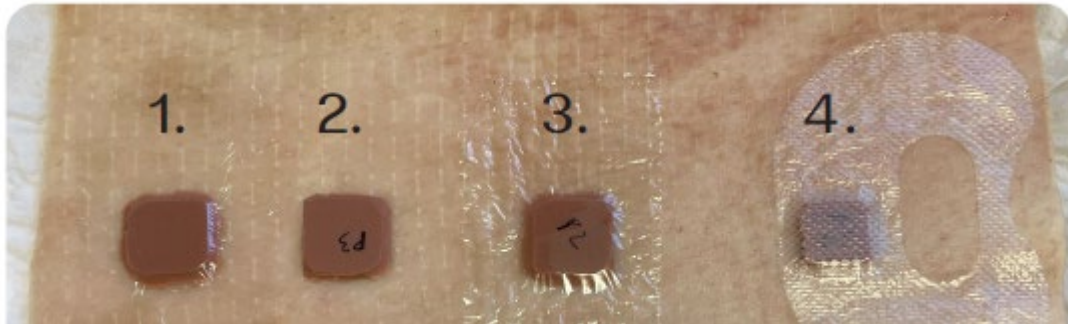


- Not easy to jump straight to immunosuppressant monitoring
- IOF-POC project not funded, because pigs are not humans
- TRL high enough to ethical approval and pre-clinical trials
- Look for other ways to build knowledge, build TRL and trust to make this investment

How do we go from concept to trial?



Penetration testing - Porcine skin



Adhesion testing - Porcine skin



10N



10,2N

Adhesion testing - Porcine skin



Ingress testing - IPX7



Adhesion testing - Human skin (sweat, running)

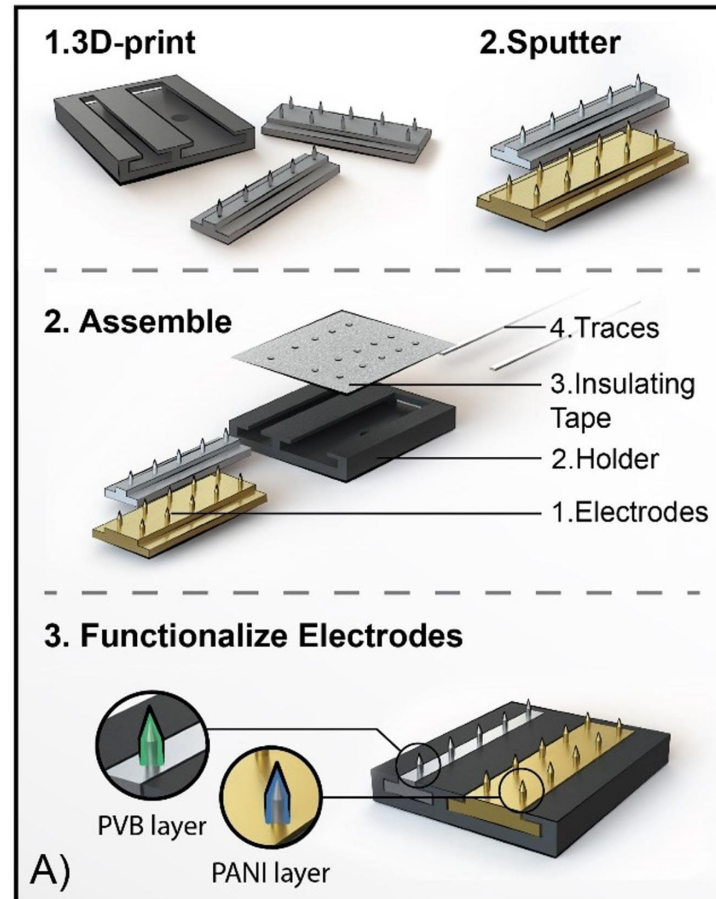
How do we go from concept to trial?

- Photopolymer resin used in the 3D printed microneedle array not biocompatible, so unsuitable material for human studies!
- Look to implement your technology on a living organism that doesn't need ethical approval and pre-clinical trials
- Recent publication with colleagues from A-PECS

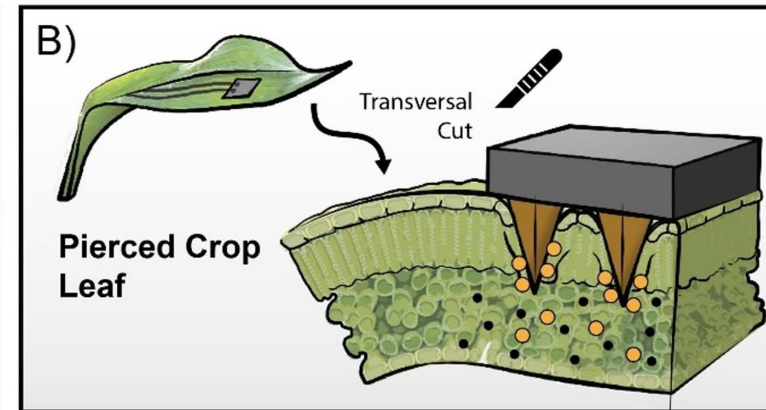
Parrilla, Marc, Annemarijn Steijlen, Robbe Kerremans, Joe Jacobs, Louise den Haan, Janne De Vreese, Yorben Van Noten Geron, Phil Clerx, Regan Watts, and Karolien De Wael. "Wearable platform based on 3D-printed solid microneedle potentiometric pH sensor for plant monitoring." *Chemical Engineering Journal* 500 (2024): 157254.

Wearable platform based on 3D-printed solid microneedle for plant monitoring

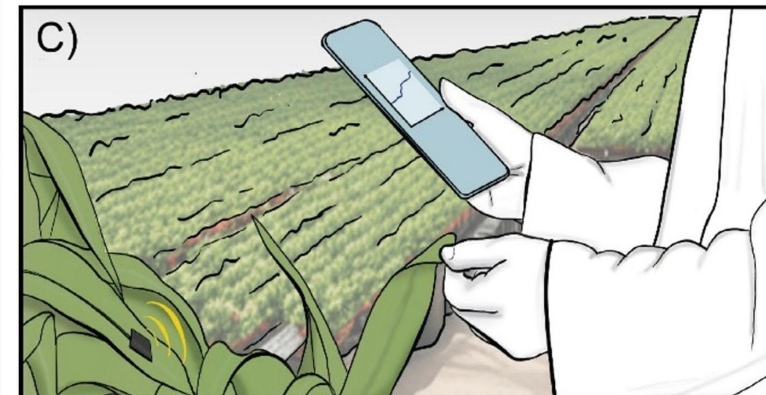
Fabrication



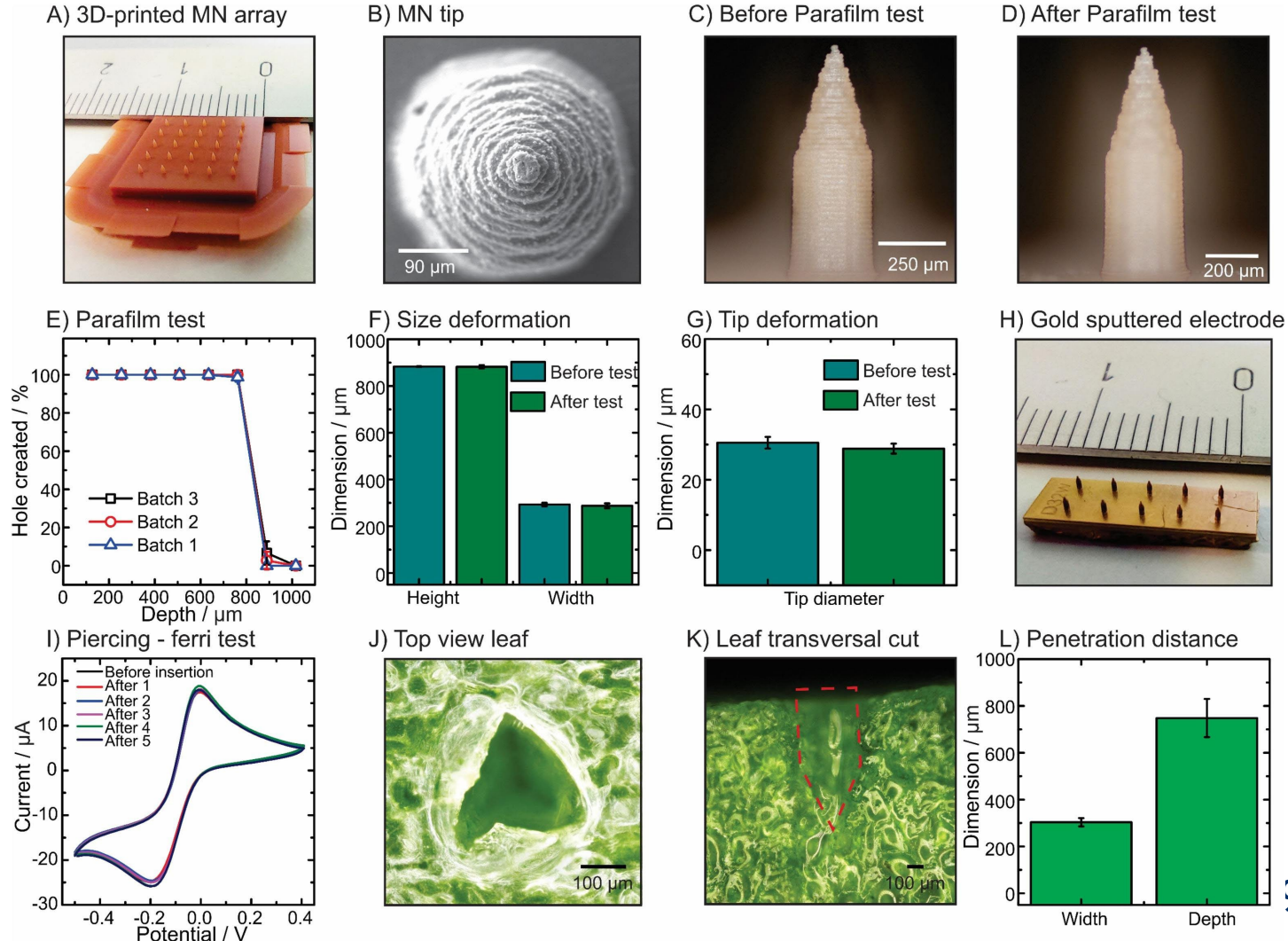
MN-Plant Insertion



Field Monitoring

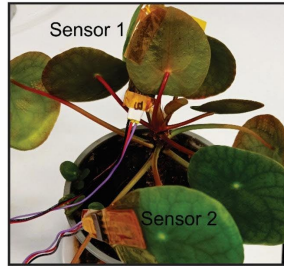


Wearable platform based on 3D-printed solid microneedle for plant monitoring

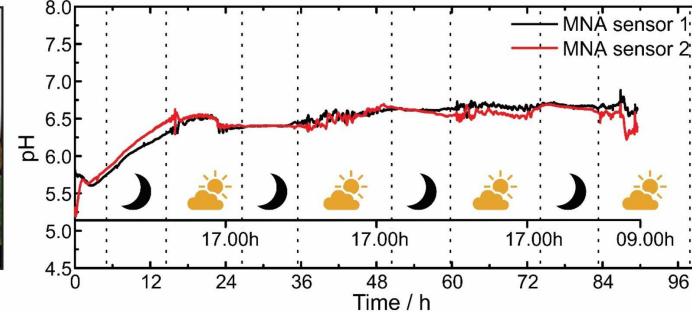


Wearable platform based on 3D-printed solid microneedle for plant monitoring

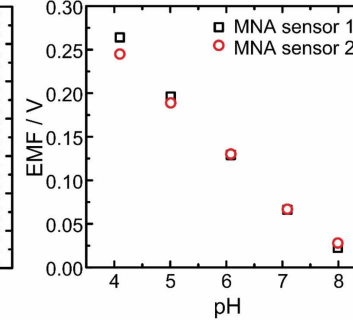
A) *Pilea peperomioides*



B) pH monitoring *Pilea peperomioides* (two MNA sensors)



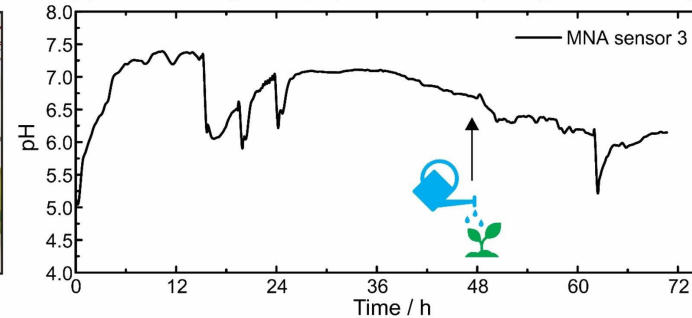
C) Calibration curve



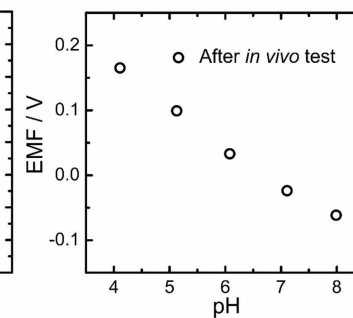
D) Drought conditions



E) pH monitoring *Pilea peperomioides* (drought conditions)



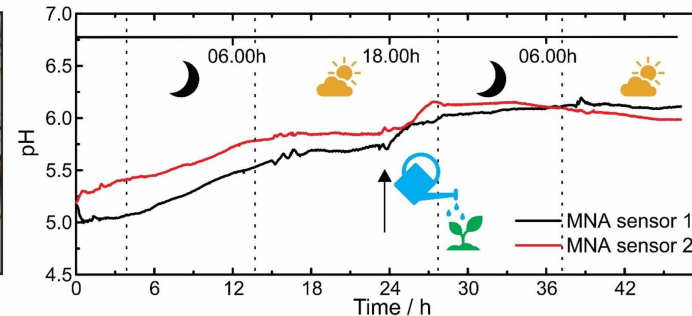
F) Calibration curve



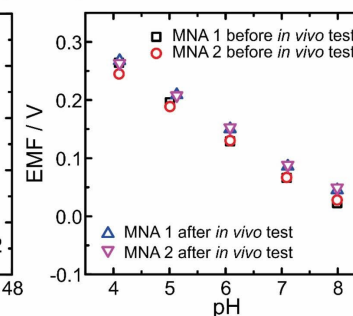
G) *Peperomia polybotrya*



H) pH monitoring *Peperomia polybotrya* (two MNA sensors)



I) Calibration curve



The elephant in the room – Clinical trials



Simon Callaerts • 1st

Director of Product Development bij Bedal

1mo • 🌐

I am proud to announce that we have achieved our full MDR certification!



Bedal

1,397 followers

1mo • 🌐

+ Follow

We're excited to announce that we've achieved full MDR certification for our FlexGRIP product line!

This milestone confirms our products meet and exceed the EU's rigorous medical device safety and quality standards.

Thanks to our amazing team that has dedicated countless hours of their expertise and persistence to reach this MDR certification!

#MDR #MedicalDevices #Healthcare #MedTech #QualityMatters
#CatheterCare #FlexGRIP

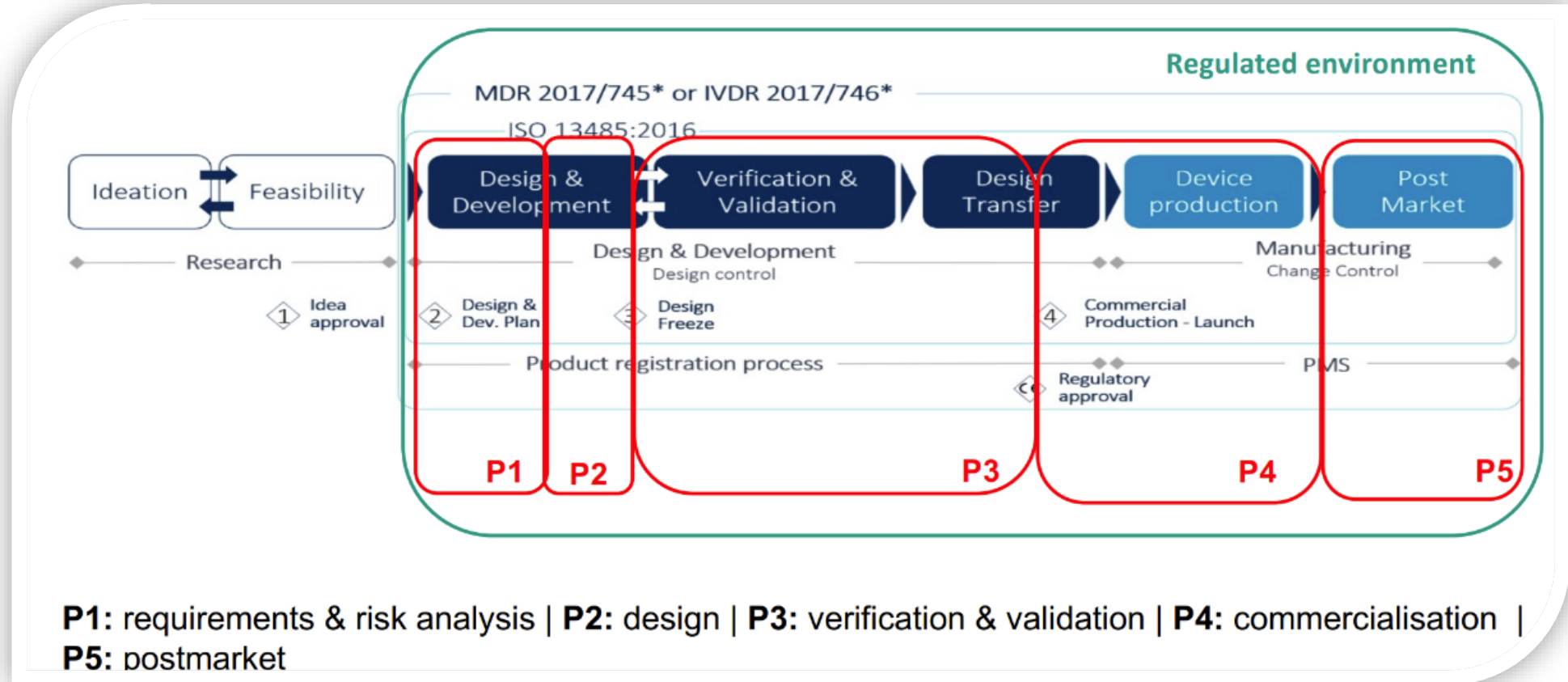


🌐 You and 41 others

2 comments



The elephant in the room – Clinical trials



- Not even at P1 - Clinical study strategy and plan
- Keep going, building out use cases

Infrastructure

Measure Lab 50 m² lab space

- Cleanroom assembly
- Microscopy and validation
- Ultrasonic welders for medical textiles
- Autoclave and glassware dishwasher for sterilization/hygiene
- New tensile testing machine (fingers crossed)
- Fabrication techniques (vacuum forming, laser cutting, CNC-milling, casting, mSLA, SLA, FFF, SLS printers)

Part 2: “Work In-Progress” Report from the Design Coal Face

with Michael Deman, Lynqo

Part 3: Antwerp Design Factory

ANTWERP
DESIGN
FACTORY



TABLE OF CONTENTS

1 Antwerp Design Factory

2 Valorisation

3 Innovation funnel

4 Collaboration possibilities

OUR MISSION

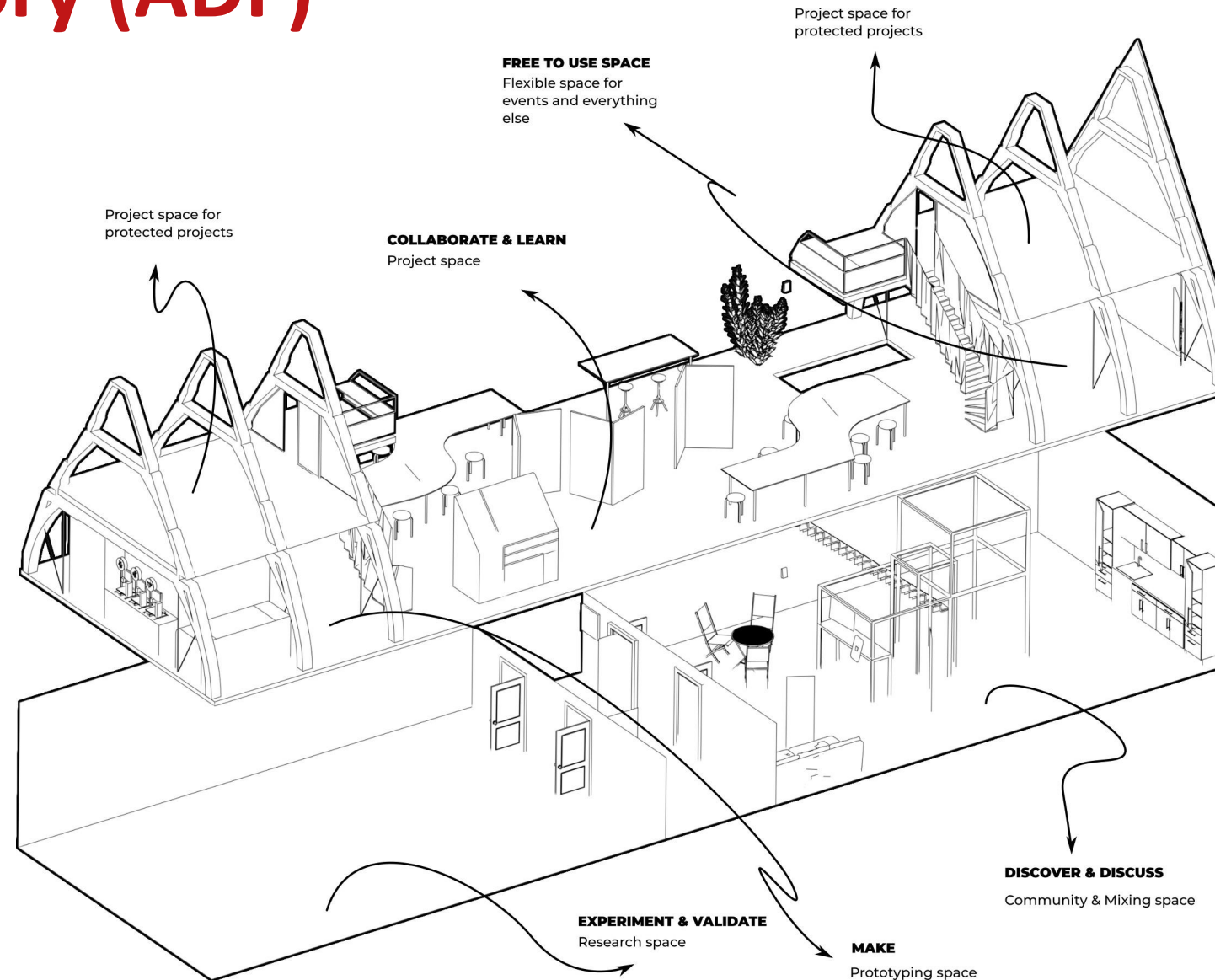
BRINGING MEANINGFUL
PRODUCTS TO THE MARKET

Antwerp Design Factory (ADF)

A **physical workspace**, where students and researchers from different faculties can **collaborate** to make **innovative** products and services

Where unique **design expertise** and **manufacturing facilities** are offered

Pre-incubation space



Antwerp Design Factory (ADF)



Svetlana
Samsonova



Florentijn
Vandecasteele



Ralv
Lundahl



Sien
Heirbaut



Alexander
Van Gastel



Jouke
Verlinden



Lukas
Van Campenhout



Stijn
Verwulgen



Regan
Watts



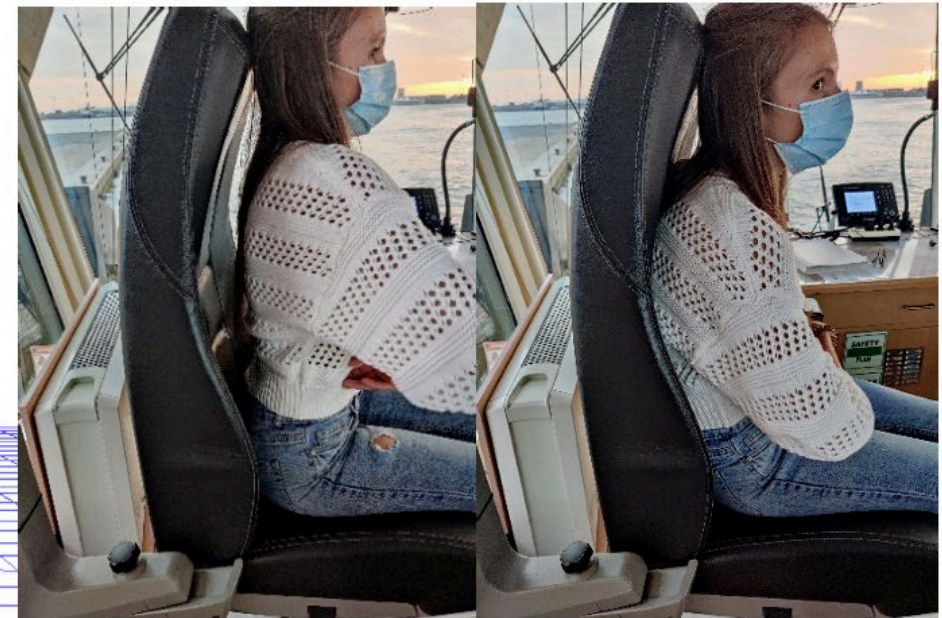
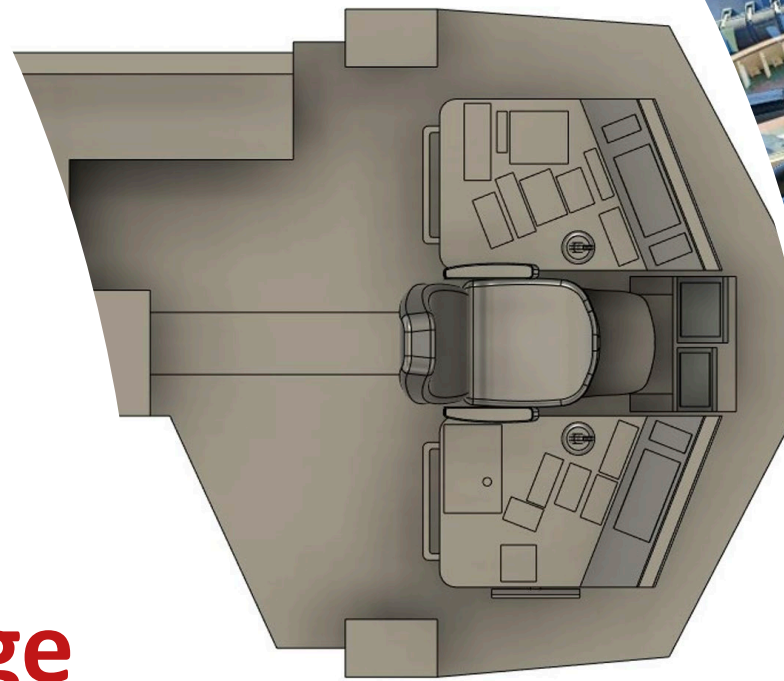
Kristof
Vaes

Smart Docking LiDAR

2023



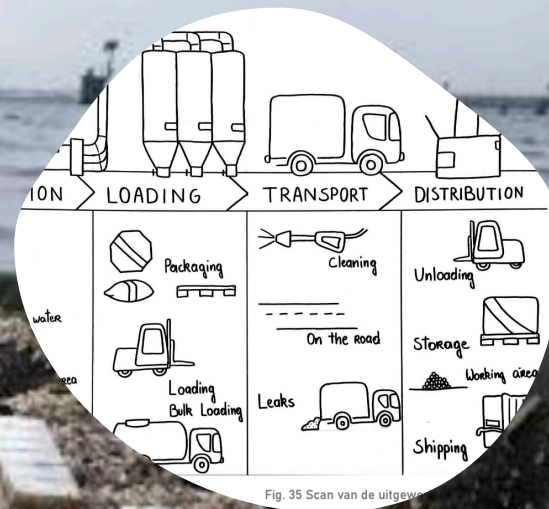
Re-Design Tugboat Bridge



2021

Zero Pellets Loss

2020



Lashing tool

2022





“To patent or not to patent,
that is the question”

-Hamlet



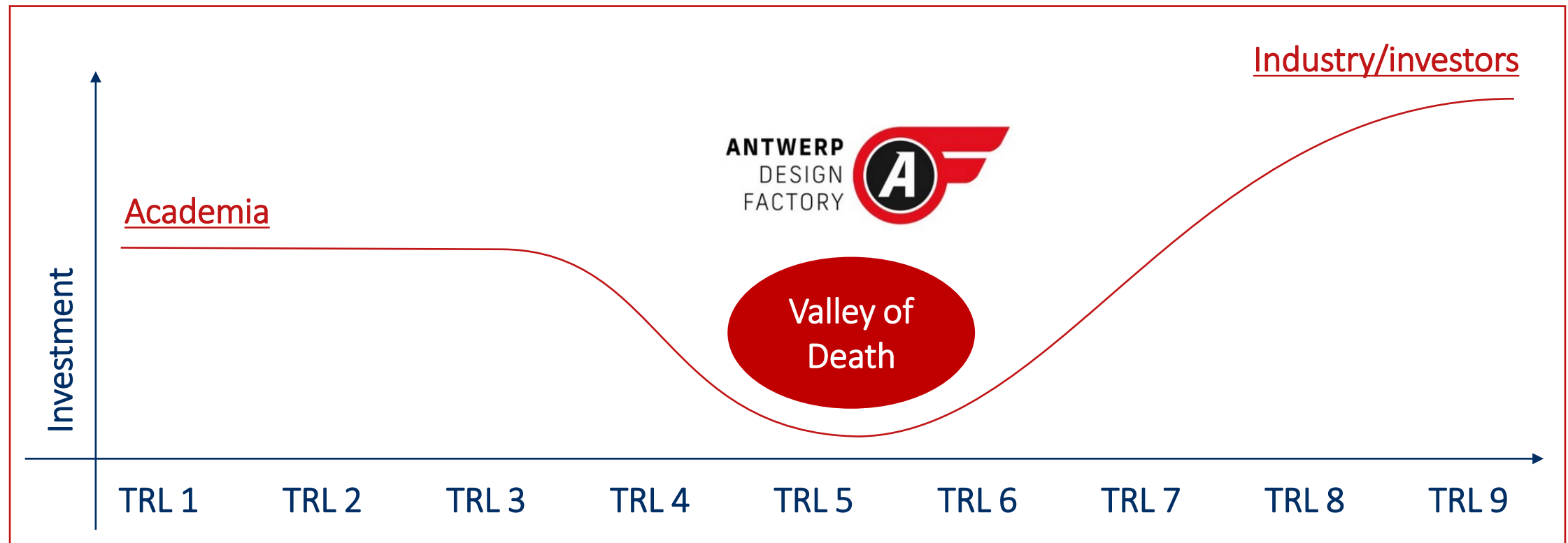
We want spin-
off's

*"Because every great idea
deserves its own spotlight!"*

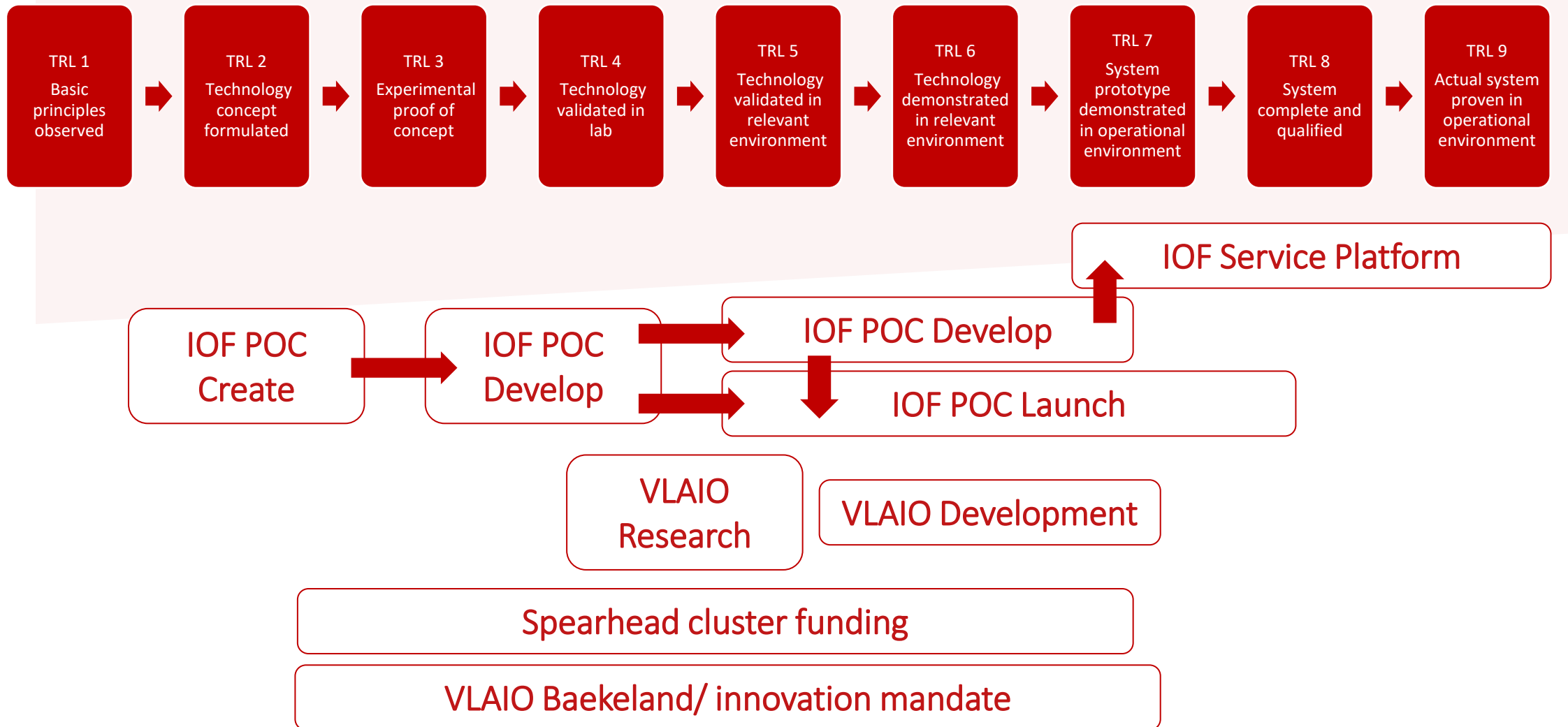
What is valorisation?



Using the knowledge from **innovative research** for the benefit of the **economy** and **society** by means of a product, process, organization or business model.

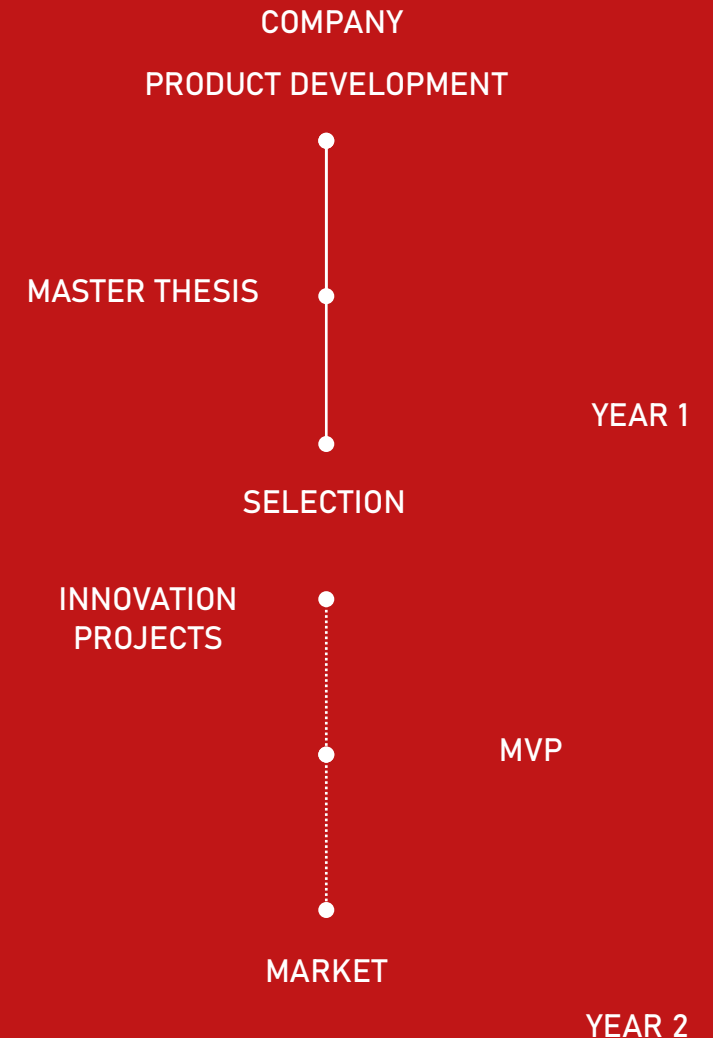


Innovation funnel & funding channels

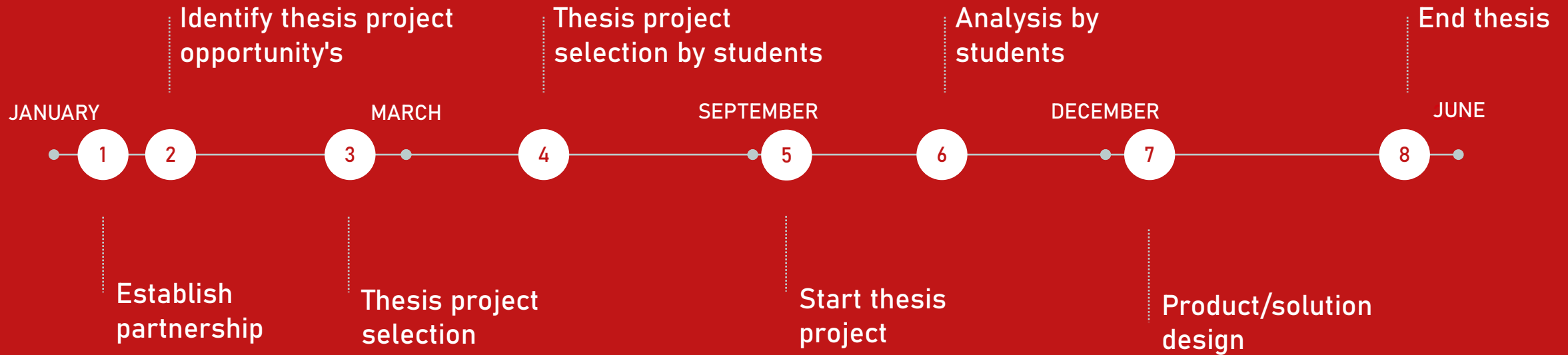


- 1 Define common goals
- 2 Partnership framework
- 3 Ideation and selection
- 4 Innovate on promising projects
- 5 Impact and commercialization

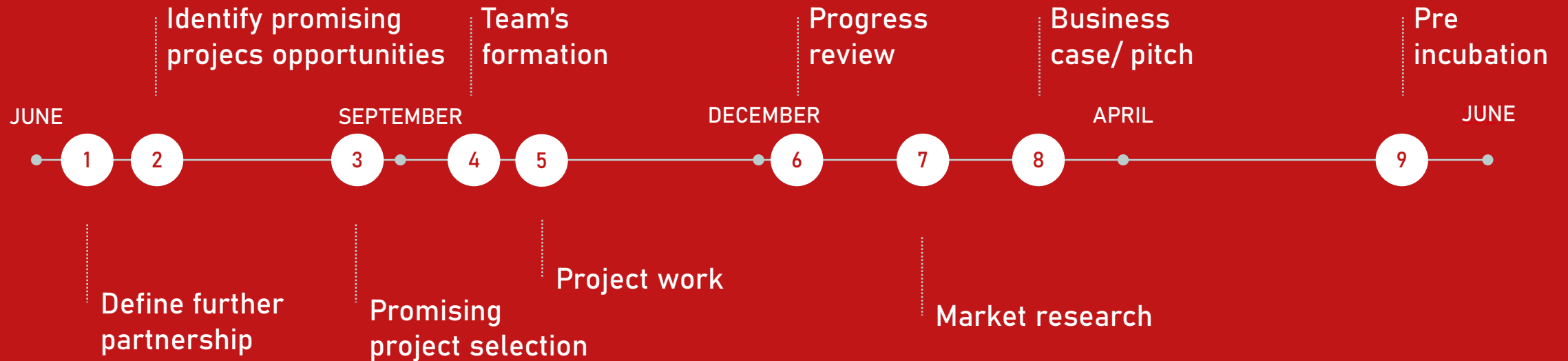
PARTNERSHIP



YEAR 1: MASTER THESIS



YEAR 2: VALORISATION



Part 4: Tour Antwerp Design Factory and School of Product Development

12 February 2026



University of Antwerp
| Product Development

Thank you for your interest!

Prof. dr. Regan Watts

First a group photo?